7 **(a)** Simplify.

$$5g - 3h - 7g + 6h$$

..... [2]

(b) $j = 4k + 7m$

Find the value of j when $k = -5$ and $m = 6$.

$j =$ ..... [2]

(c) Factorise completely.

$$14x^3 + 49x$$

..... [2]

(d) Solve.

$$8(3t - 9) = 108$$

$t =$ ..... [3]

(e) **(i)** $9^{24} \div 9^w = 9^5$

Find the value of w.

$w =$ ..... [1]

(ii) $4x^2 = 256$

Find the value of x.

$x =$ ..... [1]

(f) Ranjit's age is x years.
Suzi's age is 3 times Ranjit's age.
Juan's age is 4 years more than Suzi's age.
The total of their ages is 46 years.

Use this information to write down an equation and solve it to find the value of x.

$x =$ ..... [4]